

1 million records =>

Without Pandas	Pandas
Manual loops	Fast -> fun, methods
Complex logic	Simple Syntax.
Slow analysis.	powerful & Robust but - is operations

Key features:-

- Data Structure: Series -> one dimensional labeled array
Data frame -> Two dimensional labeled array
- Handle missing data: Built in methods for finding missing values
- Data Alignment: Automatic alignment of data during operations
- Flexible Indexing: Label based and integer-based indexing
- Data I/O: Read/write -> CSV, Excel, SQL, JSON etc.
- Powerful Grouping: groupby -> aggregations & transformations
- Time Series Support: Datetime

Used in

Data Analysis
Data Science
ML
Finance
B.A.

What is Pandas?

- Python library -> data manipulation and analysis
- Mainly used:
 - load data (CSV, Excel, SQL, JSON)
 - clean data
 - Analyze data & Transform data

Pandas work best with structured / tabular data (rows & columns)